

Souranil Das

LinkedIn: [linkedin.com/in/souranild](https://www.linkedin.com/in/souranild)

Github: github.com/souranild

Email: souranil.das@gmail.com

Mobile: +91-700-8459-203

EDUCATION

- Vellore Institute of Technology** Chennai, India
Bachelor of Technology - Electronics and Communication; GPA: 3.23/4.0
Courses: Data Structures & Algorithms, VLSI, IOT, Computer Networks 2019 - 2023

SKILLS SUMMARY

- Languages:** SystemVerilog, C/C++, Python, Bash
- Technologies:** IoT, Embedded Systems, RTL Design, Automation, JTAG, Ethernet
- Verification:** Functional Verification, Assertion-Based Verification, UVM, SystemVerilog

EXPERIENCE

- NXP Semiconductors** Bengaluru, India
Design Verification Engineer - G2 Apr 2024 – Present
 - Working** on an ethernet based multi interface automotive hub.
 - Working** on an internal tool for simplifying test running, and monitoring.
 - Successfully completed **4** tape-outs.
- NXP Semiconductors** Bengaluru, India
Design Verification Engineer - G1 Jul 2023 – Apr 2024
 - Worked** on updates on Matrix LED Controller, which provides smart automotive headlights and backlights.
 - Completed **two** incremental tape-outs.
 - Automated** and made DV work environments better by bringing in things learnt from Software Validation.
- NXP Semiconductors** Hyderabad, India
Software Validation & Automation Intern Sept 2022 – July 2023
 - Developed** tests and automation scripts for Realtime Edge IoT platform
 - Shipped 3** successful releases with fully automated regression setups
- Samsung Electronics** Bengaluru, India
Project Intern June 2021 – Jan 2022
 - Developed** an internal web application using ReactJS, Flask, and TensorFlow for image and video annotation.
 - Integrated** Advanced ML Models: Incorporated SSD, RCNN, and YOLOv3 for data testing and batch processing workflows.
- Yuan-Ze University** Taoyuan, Taiwan
IoT Research Intern June 2022 – March 2023
 - Collaborated** with YZU Taiwan students to design a Smart Hydroponic System with sensors and automation for basil growth.
 - Developed** a Smart Hydroponic System, involving hardware and software co-creation. Achieved a fully automated system for monitoring indoor organic vegetable growth, including testing various hydroponic solutions, sensor combinations and reporting plant vitals.
- IEEE Photonics** Chennai, India
Vice President July 2020 – July 2021
 - Headed** the development of the society's website and collaborating on tasks such as scheduling talks.
 - Managed** administrative functions, contributing to the smooth operation of the Society.

PROJECTS

- AI Annotation Tool for Data Testing (ReactJS, Flask, TensorFlow, SSD, RCNN, YOLOv3):** Developed a full-stack web application for image and video annotation. Integrated object detection models. Deployed internally for data testing and batch processing workflows. (Jun 2021 - Jan 2022)
- Access Control System using 8051 Microcontroller and DC Motor (Microcontroller, RFID):** Developed an RFID-based access control system using 8051 microcontroller. Enabled motor control for authorized entries, unauthorized access alerts, and real-time entry count tracking. (Feb 2021 – Jun 2021)
- Remote Video Surveillance Wireless Robot (Wi-Fi, Video Streaming, Object Detection):** Built a Wi-Fi controlled robot with live video streaming, object/person detection, and intrusion alerts. Controlled via web server and Android app for remote surveillance. (Feb 2021 – Jun 2021)
- Garduino - Automatic Garden Management with Arduino (Arduino, IoT, Wi-Fi):** Designed and built an Arduino-based smart garden system with both manual and automatic plant watering capabilities. Enabled remote monitoring and control via smartphone using Wi-Fi, providing real-time plant status updates and user interaction. (Apr 2020 – Sep 2020)

PUBLICATIONS

- **Revolutionizing Holy-Basil Cultivation with AI-Enabled Hydroponics System (IEEE Access, 2023):** Co-authored the paper published in IEEE Access, vol. 11, pp. 82624–82639. The work integrates hydroponics, IoT, and machine learning to optimize holy-basil cultivation with real-time monitoring and healthiness scoring.
- **SMADE - Smart Medical Assist Device (Springer, 2022):** Co-authored the paper published in Computer, Communication, and Signal Processing (ICCCSP 2022), IFIP AICT vol. 651. Developed a device for elderly care integrating medication scheduling and real-time vitals monitoring.

CERTIFICATIONS

- Assertion-Based Verification - Mentor Graphics, Sep 2023
- Essential SystemVerilog for UVM - Cadence, Sep 2023
- Designing and Modelling of IoT, AI & ML Systems - Arm, Aug 2022
- AWS Certified Cloud Practitioner - Amazon Web Services, Jun 2022